

Constructing Diversification Constraints

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1 Overview

Consider the function, $f : \mathbf{R}^n \rightarrow \mathbf{R}$, defined by¹

$$f(\mathbf{x}) = \sum_{i=1}^k x_{[i]} \quad (1)$$

From this we see that $f(\mathbf{x})$ is the value of the sum of the k largest values of its input $\mathbf{x}$. We are interested in optimization problems involving a vector $\mathbf{x}$ with the sum of the top values constrained by a given value; that is, such that $f(\mathbf{x}) \leq M$ for some value M . How could we do this? One way is to write down all possible combinations of k elements from $\mathbf{x}$ and write a constraint that bounds their sum to be less than or equal to M . But the number of constraints that one has to write is $\binom{n}{k}$. This becomes combinatorially large very quickly. In the next section we seek a way to represent f to reduce the number of constraints.²

This constraint often arises in the world of finance where one wants to create (or modify) a portfolio so that the new portfolio has the property that its top k securities (notionally) do not exceed some fixed percentage of its total notional.

¹For a given vector, $\mathbf{x} \in \mathbf{R}^n$, create a vector, $\mathbf{x}'$, by *stably* sorting $\mathbf{x}$ from highest to lowest. Define the notation, $x_{[i]}$ by: $x_{[i]} \equiv x'_i$. Note: For the rest of this paper we assume that $k \in \mathbf{Z}^+$ with $k \leq n$.

²A bold font is used to denote vectors. We refer to a vector of 0s as $\mathbf{0}$, and likewise we refer to a vector of 1s as $\mathbf{1}$. The mathematical expression, $\mathbf{a} \preceq \mathbf{b}$, is to be interpreted as the statement that every element of the vector $\mathbf{a}$ is less than or equal to the corresponding element in the vector $\mathbf{b}$. Similarly, the expression, $\mathbf{a} \succeq \mathbf{b}$, is the statement that each element of $\mathbf{a}$ is greater than or equal to each corresponding element of $\mathbf{b}$.

2 The Function f as an Optimization Problem

We wish to characterize the value of f on a given vector, $\mathbf{x}$, as the result of an optimization. We claim that for a given $\mathbf{x}$, $f(\mathbf{x})$ is the value of a solution to the following constrained *discrete* optimization problem:

$$\max_{\mathbf{y} \in \mathbf{Z}_2^n} \mathbf{y}^T \mathbf{x} \quad (2)$$

$$\mathbf{y}^T \mathbf{1} = k \quad (3)$$

In English this says: “Take the maximum value of all possible sums of k values from $\mathbf{x}$.” This is clearly just another way of restating $f(\mathbf{x})$. Although this optimization is succinct, there are still combinatorially many combinations to examine to find the optimal solution in this discrete setting.

claim: The discrete optimization problem is equivalent to the following continuous problem:³

$$\max_{\mathbf{y} \in \mathbf{R}^n} \mathbf{y}^T \mathbf{x} \quad (4)$$

$$\mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1} \quad (5)$$

$$\mathbf{y}^T \mathbf{1} = k \quad (6)$$

proof: Since its domain, $\mathbf{R}^n$, includes $\mathbf{Z}_2^n$, a solution to the continuous problem is at least as large as any solution of the original discrete problem. We now show the reverse, by showing that for a given optimal solution to the continuous problem, $\mathbf{y}^*$, there is a corresponding $\mathbf{y}'$ that lives in $\mathbf{Z}_2^n$, satisfying the constraint, (3), with (2) yielding the same value for its objective as the optimal value of the continuous problem.

To this end, let $\mathbf{y}^* \in \mathbf{R}^n$ be a solution to the continuous problem. If all of the values are integral, by (5), the only integral values are 0 or 1. In this case, we are done – take $\mathbf{y}'$ to be $\mathbf{y}^*$.

Otherwise, collect the components of $\mathbf{y}^*$ that are non-integer; that is, have values strictly between 0 and 1. By the constraint, (6), the sum of these non-integer components must be an integer. Consequently, there must be more than one such component. It must also be the case that amongst these non-integer components the corresponding “ $\mathbf{x}$ ” values are the same. Otherwise, if two of these components had differing “ $\mathbf{x}$ ” values, then the “ $\mathbf{y}$ ” component with

³A solution to the continuous problem exists as the supremum of a *continuous* objective function over a closed, bounded set in $\mathbf{R}^n$ – as determined by these particular constraints – is attained. Hence, we may use the term “max” rather than “supremum” for the optimal objective value.

the smaller “x” value could shift some of its value to the “y” component with the larger “x” value in such a way as to preserve the constraints, (5) and (6). But then we would have a larger objective value than the supposed maximum – contradiction. Therefore, all of the “x” components associated with non-integer “y” values must have the same value. As previously stated, the sum of these non-integer values is an integer, which by necessity is less than the number of non-integer components in $\mathbf{y}^*$. Let j denote this integer sum. Pick any j of these non-integer values, and let J be the set of their index values in $\mathbf{y}^*$. Let $\tilde{J}$ be the set of indices corresponding to the remaining non-integer values of $\mathbf{y}^*$.

Now, create a new vector $\mathbf{y}' \in \mathbf{Z}_2^n$ and set its values in the following way: Set the values of $\mathbf{y}'$ corresponding to the indices in J to be 1. Set the values of $\mathbf{y}'$ corresponding to the indices in $\tilde{J}$ to 0. Note that with this assignment, the sum of the elements of $\mathbf{y}^*$ with indices in $J \cup \tilde{J}$ is the same as the sum over $\mathbf{y}'$ with indices in $J \cup \tilde{J}$. Set the rest of the components of $\mathbf{y}'$ to the corresponding values from $\mathbf{y}^*$. By our construction, $\mathbf{y}'^T \mathbf{1} = k$. By constraint, (5), the integer values of $\mathbf{y}'$ which came from $\mathbf{y}^*$, must be either 0 or 1. It follows that $\mathbf{y}' \in \mathbf{Z}_2^n$ and satisfies the constraint of the discrete problem, (3). The objective value is also preserved: since the “x” components on $J \cup \tilde{J}$ all share a common value, x_c , the contribution to the objective from those indices equals $x_c \sum_{i \in J \cup \tilde{J}} y_i^* = x_c j$, which is exactly the contribution from $\mathbf{y}'$ on the same indices. Therefore, $\mathbf{y}'$ is in the feasible set of the discrete problem and attains the optimal value of the continuous problem; meaning, the solution to the discrete problem is at least as large as any solution to the continuous problem. Consequently, the two optimization problems are equivalent.

The continuous problem above is equivalent to the following minimization:

$$\min_{\mathbf{y} \in \mathbf{R}^n} \quad -\mathbf{y}^T \mathbf{x} \tag{7}$$

$$\mathbf{0} \preceq \mathbf{y} \preceq \mathbf{1} \tag{8}$$

$$\mathbf{y}^T \mathbf{1} = k \tag{9}$$

Treating $\mathbf{x}$ as a constant, this is a *convex* optimization problem.

3 A Dual Description of the Optimization

Since the problem described by equations (7, 8, 9), is *convex*, the value of its solution is the same as the value of its associated *dual* problem.⁴

⁴Normally one needs to show that a convex problem satisfies Slater’s condition in order to claim that it provides the same value as its associated dual. But this is not necessary when dealing with *linear* convex

To form the dual problem we need the Lagrangian, which is:⁵

$$L(\mathbf{y}, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = -\mathbf{y}^T \mathbf{x} - \boldsymbol{\lambda}_1^T \mathbf{y} + \boldsymbol{\lambda}_2^T (\mathbf{y} - \mathbf{1}) + \nu(k - \mathbf{y}^T \mathbf{1}) \quad (10)$$

Here $\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2 \in \mathbf{R}^n$ and $\nu \in \mathbf{R}$ are the *dual* variables.

Set the function g :

$$g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = \inf_{\mathbf{y}} L(\mathbf{y}, \boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) \quad (11)$$

Substituting for L this becomes

$$g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) = \inf_{\mathbf{y}} (\mathbf{y}^T (-\mathbf{x} - \boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 - \nu \mathbf{1}) - \mathbf{1}^T \boldsymbol{\lambda}_2 + \nu k) \quad (12)$$

The dual problem is then

$$\max_{\substack{\boldsymbol{\lambda}_2 \succeq \mathbf{0} \\ \boldsymbol{\lambda}_1 \succeq \mathbf{0} \\ \nu}} g(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2, \nu; \mathbf{x}) \quad (13)$$

Which is⁶

$$\max_{\substack{\boldsymbol{\lambda}_2 \succeq \mathbf{0} \\ \boldsymbol{\lambda}_1 \succeq \mathbf{0} \\ \nu}} -\mathbf{1}^T \boldsymbol{\lambda}_2 + \nu k \quad (14)$$

$$-\boldsymbol{\lambda}_1 + \boldsymbol{\lambda}_2 - \nu \mathbf{1} = \mathbf{x} \quad (15)$$

This is equivalent to⁷

$$\max_{\substack{\boldsymbol{\lambda} \succeq \mathbf{0} \\ \nu}} \nu k - \mathbf{1}^T \boldsymbol{\lambda} \quad (16)$$

$$\mathbf{x} \preceq \boldsymbol{\lambda} - \nu \mathbf{1} \quad (17)$$

problems. Why, you may ask is it *linear*? The term $-\mathbf{y}^T \mathbf{x}$ is decidedly non-linear. It is linear if we treat $\mathbf{x}$ as a constant (we also use the term parameter).

⁵For the purposes of this optimization, $\mathbf{x}$ is seen as a parameter to the optimization – essentially a constant with respect to the optimization. As such, it is customary to treat such parameters as part of a function but separate them from the “true” list of variables of the problem with a semi-colon.

⁶Note that g is $-\infty$, unless the term that $\mathbf{y}$ is “dotting” is the zero vector. Consequently, the maximum must necessarily occur where the dotting vector is zero.

⁷We can remove n variables by eliminating $\boldsymbol{\lambda}_1$ from the equations while keeping the same number of inequalities. We can do this by realizing that $\boldsymbol{\lambda}_2 - \nu \mathbf{1} = \mathbf{x} + \boldsymbol{\lambda}_1$ expresses the same information as $\boldsymbol{\lambda}_2 - \nu \mathbf{1} \succeq \mathbf{x}$. Since there is now only one $\boldsymbol{\lambda}$, we relabel $\boldsymbol{\lambda}_2$ as $\boldsymbol{\lambda}$.

Since maximizing over ν or $-\nu$ is the same – as there are no restrictions on ν – we may replace ν with $-\nu$ in the last equations giving:

$$\max_{\substack{\lambda \succeq \mathbf{0} \\ \nu}} \quad -\nu k - \mathbf{1}^T \boldsymbol{\lambda} \quad (18)$$

$$\mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \quad (19)$$

But this is the same as:

$$\min_{\lambda, \nu} \quad \nu k + \mathbf{1}^T \boldsymbol{\lambda} \quad (20)$$

$$\mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \quad (21)$$

$$\boldsymbol{\lambda} \succeq \mathbf{0} \quad (22)$$

4 Characterization of f as $O(n)$ Linear Constraints

We now apply the results of the previous sections to the following problem: Add constraints to an optimization problem that includes a *variable* $\mathbf{x} \in \mathbf{R}^n$ so that $f(\mathbf{x}) \leq M$ for some value M .

We claim that one can bound $f(\mathbf{x})$ by adding two new variables, $\boldsymbol{\lambda} \in \mathbf{R}^n$ and $\nu \in \mathbf{R}$, along with the following three constraints:

$$\nu k + \mathbf{1}^T \boldsymbol{\lambda} \leq M \quad (23)$$

$$\mathbf{x} \preceq \boldsymbol{\lambda} + \nu \mathbf{1} \quad (24)$$

$$\boldsymbol{\lambda} \succeq \mathbf{0} \quad (25)$$

Why? Because the expression $\nu k + \mathbf{1}^T \boldsymbol{\lambda}$ with the constraints (24) and (25) applied is *always* an upper bound to $f(\mathbf{x})$. Consequently, we have the inequality: $f(\mathbf{x}) \leq \nu k + \mathbf{1}^T \boldsymbol{\lambda} \leq M$. The only concern remaining is that there may be a gap between the values of $\nu k + \mathbf{1}^T \boldsymbol{\lambda}$ and $f(\mathbf{x})$ – making (23) too restrictive. But this is not the case as the minimum over all $\boldsymbol{\lambda}$ and ν (subject to (24) and (25)) is $f(\mathbf{x})$. Another way of saying this is that the parameters, $\boldsymbol{\lambda}$ and ν provide enough freedom to $\mathbf{x}$ so that $f(\mathbf{x})$ can be as close to M as one would like.

One might ask why we couldn't do exactly the same procedure with the “simpler” characterization we had with equations (7), (8), and (9), rather than going through so much effort

with a dual problem. The same number of variables and the same number of constraints are required.

We could do that; however, note that once we take the characterization of a bound on a “given” $\mathbf{x}$ and place it into an optimization problem, that “given” $\mathbf{x}$ is no longer a constant. It, itself, is part of the optimization. To enforce $f(\mathbf{x}) \leq M$ via the primal, we would have to encode the LP optimality of $\mathbf{y}^T \mathbf{x}$ against $\mathbf{x}$ as a constraint, which introduces the term $-\mathbf{y}^T \mathbf{x}$ from (7). But this term involves the (now) *variable*, $\mathbf{x}$, and the new *variable*, $\mathbf{y}$, in a decidedly *non-linear* way. In fact, the term is not even *convex*.

In contrast to this “simpler” characterization and the original naïve combinatorial approach, we have shown that by examining an associated dual problem, $f(\mathbf{x})$ can be bounded by adding $(n + 1)$ new variables and $(2n + 1)$ *linear* constraints.